

d) $\forall x \forall y \exists z (x+y=z)$
 For any two arbitrary real no.s, their sum exists and is a real no.

27) Determine the truth value of each of these statements if the domain for all variables consists of all integers.

a) $\forall n \exists m (n^2 \mid m)$. Let it be False. By defn. of negation, $\neg \forall n \exists m (n^2 \mid m)$ is True. By De Morgan's law of quantifiers, $\exists n \neg \exists m (n^2 \mid m)$ is True. By De Morgan's law of quantifiers, $\exists n \forall m \neg (n^2 \mid m)$ is True. $\therefore \exists n \forall m (n^2 \nmid m)$ is True. By defn. of existential quantifier, there exists atleast one integer n where $\forall m (n^2 \nmid m)$ is True. By defn. of universal quantifier, for all integers y , $(n^2 \nmid y)$ is True. $\therefore n^2 \nmid n^2 + 1$ is True $\Rightarrow \boxed{0 \nmid 1}$. This is False and a contradiction.

$\therefore \forall n \exists m (n^2 \mid m)$ is True

b) $\exists n \forall m (n \mid m^2)$. Let it be False. By defn. of negation, $\neg \exists n \forall m (n \mid m^2)$ is True. By De Morgan's law of quantifiers, $\forall n \neg \forall m (n \mid m^2)$ is True. By De Morgan's law of quantifiers, $\forall n \exists m \neg (n \mid m^2)$ is True. $\therefore \forall n \exists m (n \nmid m^2)$ is True. By defn. of universal quantifier, for all integers a in the domain, $\exists m (a \nmid m^2)$. By defn. of existential quantifier, for all integers a in the domain, there exists some integer b (not constant) such that $a \nmid b^2$. We know the fact that $b^2 \nmid 0$ for all $b \in \mathbb{R}$. $\therefore \boxed{a \nmid 0}$. But this is a contradiction for all $a < 0$, i.e. negative no.s.

$\therefore \exists n \forall m (n \mid m^2)$ is True

c) $\forall n \exists m (n+m=0)$. We know, $\forall n n+(-n)=0$. $(-n) \in \mathbb{R}$. By defn. of existential quantifier, $\forall n \exists m (n+m=0)$ is True

d) $\exists n \forall m (nm=m)$. We know, $\forall m 1 \cdot m = m$. By defn. of existential quantifier, $\exists n \forall m (nm=m)$ is True.

e) $\exists n \exists m (n^2+m^2=5)$. We know, $1^2+2^2=5$. By defn. of existential quantifier, $\exists n \exists m (n^2+m^2=5)$ is True.